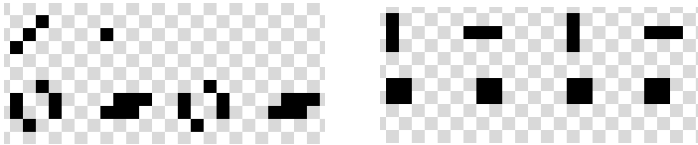


Game of Life

Game of Life rules: a cell will stay alive if 2-3 neighbors are alive otherwise it dies. A dead cell can become alive if exactly 3 neighbors are alive. This update happens on every positive edge of the universal clock.

Fun fact: it's possible to create logic gates inside the game of life! It is also possible to recreate the Game of Life within itself!!

Examples of creations in game life and how they move in 4 clock cycles:



The Game of Life has two phases: setting and running.

In setting, the user will press the keys (key[3]: Left, key[2]: Right, key[1]: Up, key[0]: Down) to direct each cell in an 8x8 grid. If they press a key 8 times it will wrap around.

When they're ready to plant the cell, they turn on SW[8]. This process will repeat until the user is ready to enter the running phase. The user can turn on SW[7]; this instantiation is called the Tick. Turn off SW[7] to start the running phase. It will keep running on the LED 8x8 until the user turns on SW[0] (reset). From the reset stage, to play again, they can press SW[9] to begin setting the field again.

SW[9]: Set

SW[8]: Plant Cell

SW[7]: Initialize Run

SW[0]: Reset

KEY[3]: Left

KEY[2]: Right

KEY[1]: Up

KEY[0]: Down

LED: 8x8 grid.

In order to mimic the infinite nature of the board (as the game is supposed to be played); any LED on the border will have neighbors from the opposite border, corner LEDs have neighbors from all borders.

Module Diagram:

